1

Dental hard tissue pathologies, aetiology, and their clinical manifestations

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1.1 Introduction: why practise minimally invasive (tooth-preserving) operative dentistry?

Minimally invasive operative dentistry is that aspect of restorative dentistry which repairs and/or restores damaged and defective tooth structure directly in order to maintain pulp vitality, function, and aesthetics (see Figure 1.1). The primary goal is to respect tooth structure during this process, retaining viable and biologically repairable tissues to maintain tooth vitality for as long as possible. The hard tissue damage or defects can be caused by one or more of the following:

- caries
- tooth wear
- trauma
- developmental conditions.

Minimum intervention oral healthcare is that approach to patient management where the oral healthcare team (comprised of the dentist, nurses, oral health educators, hygienists, therapists, technicians, reception staff, and practice managers), led by the dentist, act as one to provide individualized patient-centred care and advice to encourage the patient to take responsibility for and maintain their own oral health. Minimum intervention care revolves around methods of detection/diagnosis/risk assessment of oral disease, non-operative control/prevention of these conditions, minimally invasive operative repair of tissue damage, and review/maintenance/recall of the patient and the advice/care offered by the dentist/team (see Figure 1.1). The process of care planning involves the patient, including disease prevention by behaviour change and adherence, not just listing those operative procedures offered to restore damaged or defective teeth in isolation. It must be understood from the outset that even though minimally invasive operative dentistry has a pivotal role in the 'surgical' repair of damaged teeth, it alone does not provide the actual cure for dental disease—please understand that '*drilling and filling teeth does not cure caries!*' The following sections will provide an overview of the four conditions mentioned previously with respect to their aetiology, histopathology, and microbiology where relevant. An attempt will be made to relate these features to the clinical manifestations of each condition, namely carious lesions and tooth-wear lesions.

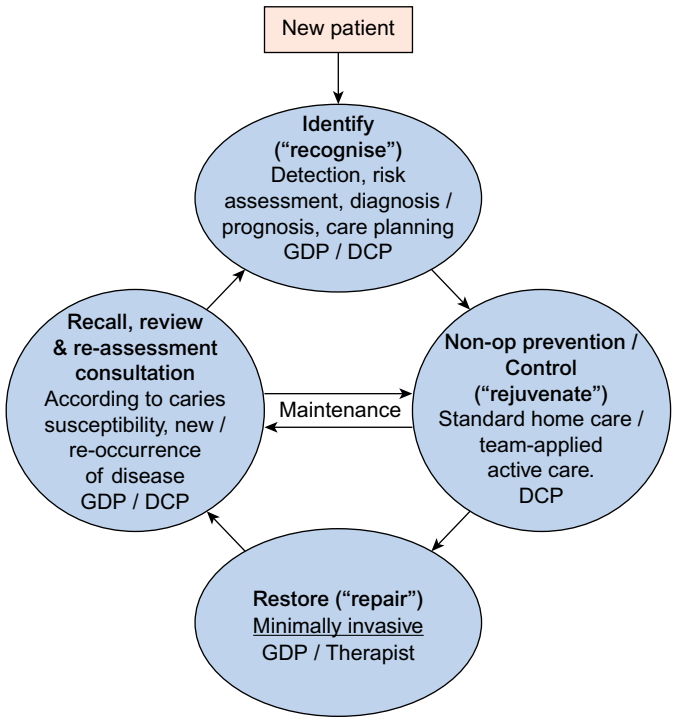


Figure 1.1 The patient-centred minimum intervention care cycle showing the four interlinking phases of patient assessment/diagnosis, non-operative prevention of lesions/control of disease, *minimally invasive* operative intervention, and review (recall). The arrows indicate the direction of patient flow through this cycle, and within each bubble an indication is given of the members of the dental team who might be included (GDP, general dental practitioner; DCP, dental care professional, including oral health educator-trained nurses, hygienists, therapists, practice managers, and reception staff).

1.2 Dental caries

1.2.1 What is it?

'A reversible (in its earliest stages) but progressive disease of the dental hard tissues, instigated by the action of bacteria upon fermentable carbohydrates in the plaque biofilm on tooth surfaces, leading to bacterially generated acid demineralization and ultimately proteolytic destruction of the organic component of the dental tissues.'

1.2.2 Terminology

Primary caries is the developing pathological biochemical process and physical lesion occurring on a previously sound tooth surface.

Root caries is primary caries on an exposed root surface (often after gingival recession has occurred), penetrating more easily into the exposed dentine. The pathological biochemical process for both primary and root caries is the same (see Figures 1.2 and 1.3).

Recurrent (secondary) caries is primary caries occurring at the margin of a failing restoration. An alternative definition of this is 'caries associated with restorations/sealants (CARS).' The aetiology is the same—metabolic activity in the stagnant plaque biofilm.

Residual caries is a term describing that portion of caries-affected, demineralized tissue retained purposely after minimally invasive cavity preparation, which is then sealed over and restored (see later).



Figure 1.2 Slowly progressing root surface lesions with dark, leathery dentine surfaces and some plaque deposits. This is the mouth of a 70-year-old patient with a dry mouth (xerostomia, secondary Sjögren's syndrome) and rheumatoid arthritis, making oral hygiene difficult due to impaired toothbrush manipulation and painful mucosae.

1.2.3 Caries: the process and the lesion

The caries process

The caries process originates as metabolic activity in the plaque biofilm resident on the tooth surface. This biofilm begins to form just a few minutes after the tooth surface has been brushed, and is adsorbed initially as the *acquired pellicle* containing an admix of salivary proteins and glycoproteins. Within a short time, oral bacteria colonize the pellicle, thus forming the dental plaque biofilm, associated closely with bacterial extracellular polysaccharides and salivary proteins. The increased density of this developing biofilm, changing bacterial population, pH, and oxygen tension all combine to create a cariogenic environment on the tooth surface. This ubiquitous natural metabolic process cannot be prevented. However, disease progression can be *controlled* by the patient, with the help of the dentist and their oral healthcare team, so that a clinically visible enamel lesion never forms. The de- and remineralization metabolic processes can be modified, particularly by regular disturbance of the biofilm with a toothbrush and fluoride toothpaste. If the biofilm is partially or totally removed at regular close intervals, mineral loss may be stopped or even reversed towards mineral gain (especially in early or *incipient* lesions). The fluoride in toothpaste delays lesion progression primarily by inhibiting demineralization and encouraging remineralization processes.

The carious lesion

The carious lesion forms as a direct consequence of the metabolic activity in the biofilm on the tooth surface (i.e. the caries process). If factors tip the demineralization/remineralization balance towards demineralization (plaque, diet, salivary factors, mineral ion concentrations, and time), the histological stages of progressive lesion formation leading to cavitation can eventually be detected clinically and dealt with accordingly.

1.2.4 Aetiology of the caries process

Occurring in the plaque biofilm, the main factors that interact in the aetiology of the carious process are as follows:

- *Bacteria*: with colonization within the plaque biofilm, several hundred different species exist within a complex ecology, dependent on the age and relative stagnancy of the plaque on the tooth surface. *Streptococcus mutans*, classically thought to be the primary causative bacterial species, is now considered to have an associative role in the



Figure 1.3 **(a)** An active root caries lesion with overlying plaque deposit in an area of stagnation alongside the margins of a partial denture. The buccal cervical abrasion cavity has been caused by excessive toothbrushing. **(b)** A stagnant plaque biofilm present on the proximal root surface, which when removed **(c)** reveals an active root caries lesion. (Images (b) and (c) courtesy of L. Mackenzie.)

Q1.1: What differences in clinical appearance are there between the coronal and root surface caries lesion in Figure 1.3 (c), and how may they relate to lesion activity?

caries process, and may act as a potential microbiological marker for caries. *Lactobacillus* and *Bifidobacterium* species have been shown to be significant in the caries process, and it is likely that species interaction within the biofilm will instigate and allow the carious lesion to progress.

- *Susceptible tooth surfaces* (see Chapter 2): carious lesions occur on tooth surfaces that have accumulated plaque, stagnating for a prolonged period of time, which may include the following:
  - The depths of pits and fissures on posterior occlusal/buccal surfaces of those teeth that the patient cannot clean effectively with a toothbrush. These areas on newly erupting molars are particularly susceptible to carious attack.
  - Proximal surfaces (mesial and distal) *cervical* to the contact points of adjacent teeth (where the patient may not floss regularly, or at all). These surfaces of particularly imbricated (crowded) teeth can be more susceptible due to the lack of access for oral hygiene aids.
  - Smooth surfaces adjacent to the gingival margin (again an area that the patient may often miss with their toothbrush), especially of those teeth that are imbricated, rotated or in-standing.
  - The ledged/overhanging/defective margins of restorations (a plaque trap created, often not evident to the patient, and inaccessible to a toothbrush or floss) (see Figure 1.4).
- *Fermentable carbohydrates*: plaque bacteria are capable of metabolizing certain dietary carbohydrates (including sucrose and glucose), producing various organic acids (lactic, acetic, and propionic acids) at the tooth surface, causing plaque pH to fall within 1–3 minutes, and initiating demineralization if the pH drops to below 5.5 (critical pH of enamel). The pH can take up to 60 minutes to climb back to normal levels, this normalization being aided by the protective buffering capacity of saliva (pH 7.0; see Figure 1.5). This demineralization/remineralization cycle occurs continuously at any tooth surface, all the time.
- *Time*: even though the drop in pH commences rapidly, sufficient time is required for the plaque biofilm to produce a *net* mineral loss equating to histological hard tissue damage at the tooth surface.



Figure 1.4 Caries at the margin of the failed dental amalgam restoration (white circle) on the occlusal surface of UL6.

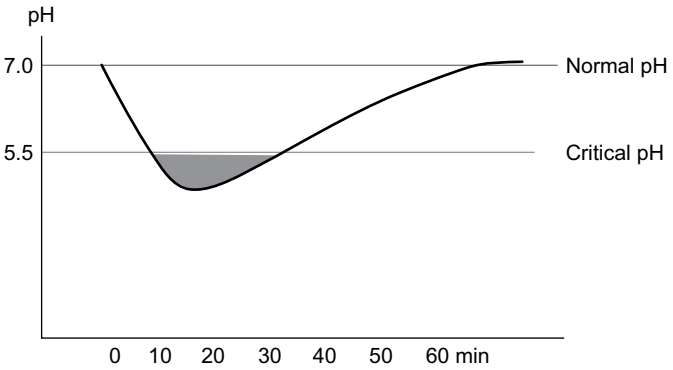


Figure 1.5 The Stephan curve, showing the changes in plaque pH over time after an oral glucose rinse at time 0 minutes. The critical pH of enamel (5.5) is that below which the hydroxyapatite crystals begin to dissociate into their constituent ions. Note that the critical pH varies, depending on an inversely proportional correlation with the concentration of available calcium and phosphate ions in the plaque biofilm fluid at the tooth surface. The grey-shaded portion of the graph indicates the 20-minute period in which the tooth surface is under threat of mineral loss. The critical pH of dentine is 6.2, which again is not fixed.

The four direct causes previously discussed can be affected or modified by several other indirect patient factors to affect ultimately the disease pattern experienced by each individual patient. These determinants include the patient's:

- income (the cost of dental care)
- knowledge about their own oral health
- attitudes to healthcare (general and oral)
- social class
- behaviour
- education.

The relative importance of these factors can be determined during verbal history taking (anamnesis) and oral examination (see Chapter 2), and helps to form the basis for determining the individual's risk and likelihood of developing caries in the future—the caries risk assessment/likelihood analysis (see Chapter 3). It is these factors in combination with the collective skills of the oral healthcare team that will help the patient to overcome or 'cure' their condition of dental caries.

1.2.5 Speed and severity of the caries process

The caries process in the normal oral environment, whose metabolic activity is tipped in favour of demineralization, will take several weeks to become detectable clinically as a lesion with signs and symptoms. This is because the overall process, with its continuously fluctuating metabolic balance at the ionic level, is relatively slow and can be moderated by oral hygiene techniques, dietary modification, and the use of fluoride or other agents. The presence of saliva, with its capacity to buffer plaque acids, provide a source of remineralizing calcium and phosphate ions to the tooth/lesion surface, remove food debris, and lubricate/protect tooth surfaces, also helps.



Figure 1.6 Early childhood caries affecting deciduous anterior teeth.

Q1.2: What habit(s) may have contributed to this pattern of disease?

'Rampant' caries

However, clinical scenarios exist where the process is accelerated and many biochemically active lesions form rapidly, often involving surfaces of teeth ordinarily expected to be caries-free—described classically as 'rampant' caries. This condition affects the primary dentition, where it is now more appropriately termed *early childhood caries* (see Figure 1.5), teenagers, or young adults with a highly cariogenic diet (frequent sugar episodes; see Figure 1.6) and/or addicted to recreational drugs, or adult patients with a dry mouth (xerostomia) (see Figure 1.7). Radiation to the region of the salivary glands, used in the treatment of an orofacial malignant growth, and Sjögren's syndrome, an autoimmune condition that may involve the salivary glands, are the most common causes of severe xerostomia. In addition, a large number of therapeutic drugs, such as antidepressants, tranquillizers, antihypertensive drugs, and diuretics can retard salivary flow and affect its quality, especially when taken together (polypharmacy).

Arrested caries

In distinct contrast to rampant caries, the term *arrested caries* describes those lesions that have stopped progressing and are inactive metabolically. It is observed when factors in the oral environment have changed from conditions predisposing to caries to conditions that tend to slow, or even reverse, lesion progression. These 'arrested' lesions often have a dark, hard, shiny exposed dentine surface (see Figure 1.8 and Table 1.1).



Figure 1.7 Rampant caries in an adult patient with cavities affecting sites not normally associated with caries due to their accessibility for adequate oral hygiene.

Q1.3: What aetiological factors may have contributed to this pattern of disease?



Figure 1.8 A hard, shiny, and stained arrested root surface lesion is present on the buccal cervical aspect of the LR4. However, the lesion on the LR3 has a matte, stippled surface appearance indicative of lesion activity. (Courtesy of L. Mackenzie.)

Q1.4: Why might the lesion on the LR3 appear active whereas the adjacent lesion on the LR4 in the same patient appears arrested?

1.2.6 The carious lesion

Having summarized the caries process as an ongoing metabolic demineralization/remineralization balance occurring at the interface between the plaque biofilm and the tooth surface, it is important to understand that the resulting carious lesion is a progressive alteration and destruction of the hard tissues (mineral and organic matrix) from the enamel surface through to the pulp. While the lesion is still within enamel, it can be arrested and possibly reversed with net mineral gain in its earliest stages. Once it is into dentine, the process can be made inactive (arrested), but if proteolytic destruction of the organic collagen matrix has occurred extensively, this cannot be reversed as observed histologically. This section will take the reader through key features of the histological and clinical development of a lesion from its earliest enamel stages through to cavitation into the pulp.

An understanding of the basic histological features of healthy enamel and dentine is an essential prerequisite for appreciating the changes

Table 1.1 Summary of differences in physical characteristics between active and inactive (arrested) carious lesions in enamel or dentine

	Physical characteristics of the carious lesion	
	Active	**Inactive (arrested)**
Enamel	Surface of enamel is white/yellow; opaque with loss of lustre; feels rough when the tip of the ball-ended probe is moved gently across the surface. Lesion is in a plaque stagnation area (i.e. pits and fissures), or near the gingival and proximal surface below the contact point. Lesion covered by plaque biofilm prior to examination (WSL)	Surface of enamel is whitish, brownish, or black. Enamel is shiny and feels hard and smooth when the tip of the ball-ended probe is moved gently across the surface. For smooth surfaces, the lesion is typically located at some distance from the gingival margin. Lesion not covered by plaque prior to examination (BSL)
Dentine	Dentine appears moist and matte; feels rough, soft, and wet or leathery on gentle probing	Dentine appears shiny and hard, and is scratchy on gentle probing

WSL, white spot lesion; BSL, brown spot lesion.

that occur within the lesion, and an outline of these is presented in Table 7.1 in Chapter 7. Further information can be obtained from the suggested further reading at the end of this chapter. The relationship between lesion histology and clinical appearance has been used in a caries detection and assessment system which is outlined and discussed in Chapter 2 (see Table 2.3).

Within enamel

Plaque-acid demineralization causes porosities to form within the prism structure, initially beneath the outer surface of enamel: this is termed *subsurface demineralization*. The developing pore volumes through the depth of the enamel lesion, caused by a longer exposure to reduced pH, have been measured using polarized light microscopy (outermost surface zone (< 1% pore volume), body (5–25%), dark (2–4%), and innermost translucent zone (1%); see Figure 1.9).

The existence of the enamel lesion surface zone may be due to increased extrinsic fluoride ion deposition in this area, or as a consequence of remineralization metabolism of the biofilm on the tooth surface. It is essential that this intact surface is not cavitated iatrogenically (i.e. a hole created by a dentist/therapist sticking a sharp dental probe/explorer into the lesion surface; see Chapter 2, Figure 2.7), as it still has the potential to heal if the biofilm can be regularly and effectively removed by the patient, and remineralizing solutions and/or toothpastes containing higher concentrations of calcium and phosphate ions used periodically (see Chapter 4, Section 4.2.3).

Histologically, smooth surface lesions have a cross-sectional shape of an inverted cone (widest superficially, with the apex towards the

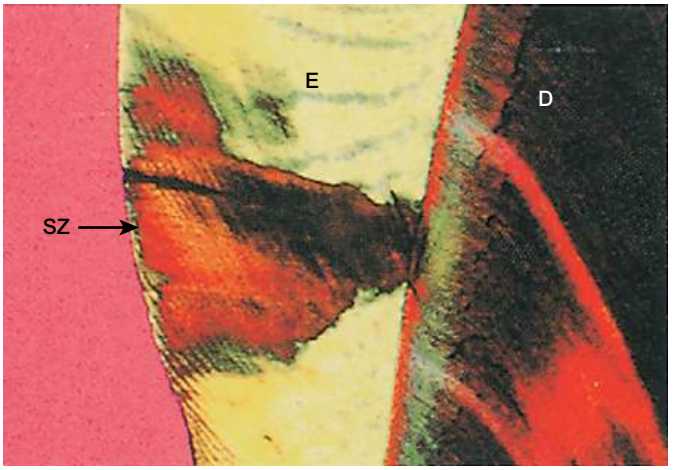


Figure 1.9 Longitudinal ground section through a carious lesion on a smooth surface (polarized light and water; E, enamel; D, dentine). The enamel lesion is shaped as an inverted cone, widest at the tooth surface, narrowing towards the enamel–dentine junction, with a relatively intact surface zone (SZ).

Q1.5: What ions have contributed to the creation of the intact surface zone and where have they come from?

enamel–dentine junction (EDJ); see Figure 1.10). Fissure lesions can be considered to take the form of two adjacent smooth surface lesions (see Figure 1.11).

- *Clinical manifestations* (see Table 1.1): the active white spot lesion (WSL) is initially smooth, frosty white/opaque, and non-cavitated



Figure 1.10 Active white spot enamel lesion on the mid-buccal of LL7 (circled). This more developed lesion has a rough surface, acting as a plaque trap.

Q1.6: What features of the lesion in Figure 1.10 will help the dentist to conclude that it is active, how might these be detected, and how might the patient be managed?

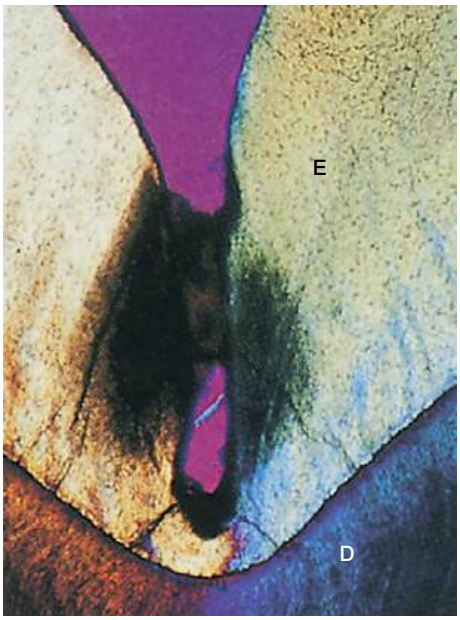


Figure 1.11 A longitudinal ground section (polarized light with water) through an occlusal fissure showing an enamel lesion forming on the two adjacent walls of the fissure (dark regions; E, enamel; D, dentine).

Q1.7: How might this scenario be managed in a high caries risk patient?

clinically (see Figure 1.12). This can be detected more easily if the tooth surface is air-dried for a few seconds using a 3-1 air/water syringe. As the lesion develops over time, it becomes somewhat chalky, eventually becoming roughened or micro-cavitated (roughness can be detected by gently running a rounded/ball-ended probe *across* the lesion surface). This surface irregularity can encourage further plaque deposition (see Figure 1.10). There are no symptoms at this stage, but reactions in the dentine–pulp complex may be mediated by cytokines and bacterial breakdown products within the dentine matrix and tubules (see later).



Figure 1.12 Early white spot enamel lesions on the cervical-gingival margins of both mandibular left molars (circled).

- If plaque is removed, these lesions can arrest and porosities can be eliminated by abrasive toothbrushing/tooth wear, resulting in the hard smooth shiny surfaces of arrested lesions. Porosities may also be filled with deposited mineral and dietary molecules causing staining (e.g. tannins), which may be trapped within the mineral lattice. This process creates an arrested brown spot lesion (BSL) with a hard shiny overlying smooth surface (see Figure 1.13b).

The lesion at the EDJ or the amelo–dentinal junction (ADJ)

Histologically, the caries process may reach dentine before clinical cavitation is detectable (a closed lesion: mICDAS score 2; see Chapter 2). Defence reactions in the dentine–pulp complex are stimulated at this stage, with evidence of translucent dentine at the advancing lesion boundary and tertiary dentine deposition at the dentine–pulp interface beneath the advancing lesion (see later). Again, symptoms are unlikely at this stage of lesion development.

The lesion extends in dentine, immediately subjacent to the EDJ (see Figure 1.13a), its lateral extension coinciding with the spread of the overlying enamel lesion at the surface of the tooth, which in turn is dependent on the extent of the resident plaque biofilm at the tooth surface. Relative hypomineralization of this histological mantle dentine zone, greater side-branching of dentine tubules, or defects within the enamel/dentine interface may also contribute to this spread laterally.

The lesion may also penetrate along the dentine tubules towards the pulp.

Within dentine

Once the lesion has spread histologically (and radiographically) further towards the middle third of dentine, it is often cavitated (open) clinically on both occlusal and smooth surfaces, with plaque now able to accumulate on the exposed dentine surface. The further spread of the lesion will undermine the overlying enamel, creating an associated visible grey shadow/opacity, which becomes brittle and prone to fracture under occlusal loading. This undermined and unsupported enamel may need to be removed during cavity preparation if the damage requires operative repair (see Chapter 5, Section 5.9.3).

The patient may begin to experience initial symptoms of acute pulpitis—a poorly localized sharp pain of a few seconds' duration stimulated by hot, cold, or sweet stimuli (see Chapter 3). The histological components of carious dentine to be considered are the mineral, collagen, bacterial penetration, and tubule structure. Both degenerative and reparative processes act on these simultaneously in different parts of the lesion. The histological changes of the carious dentine biomass through its depth (from EDJ to pulp) are described in the following bullet points, but note that these descriptive zones are not separate biological entities, but blend into one another without clear detectable boundaries (see Figure 1.14).

- *Caries-infected dentine* (zone 1 in Figure 1.14): the outermost, superficial, irreparable, necrotic zone of destruction, often distinguished clinically as a dark brown, soft, wet, 'mushy' layer.
  - Mineral component has dissociated extensively due to bacterial acid attack.

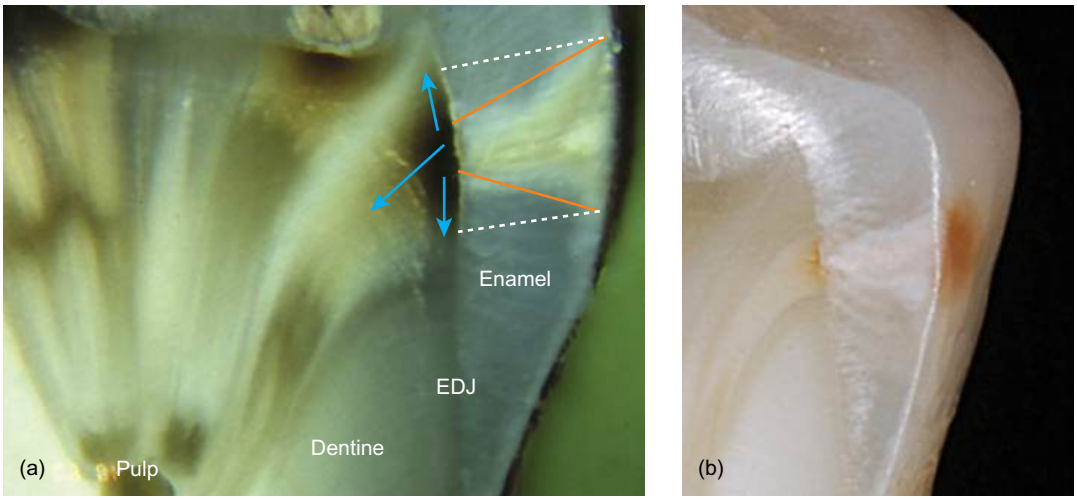


Figure 1.13 **(a)** A mesio-distal section through a carious tooth highlighting a proximal lesion. The red lines outline the 'inverted cone' cross-sectional histological shape of the enamel lesion, and the blue arrows indicate the direction of spread of the lesion having crossed the enamel–dentine junction (EDJ) into dentine. The white dotted lines show how the extent of the spread of the dentine lesion subjacent to the EDJ is associated with the same lateral extent of the enamel lesion on the tooth surface, both governed by the presence of the plaque biofilm at the tooth surface. **(b)** The surface lesion shown is an arrested brown spot lesion, its boundaries clearly aligned with the inverted cone of the enamel lesion beneath. (Figure 1.13 (b) courtesy of L. Mackenzie.)

Q1.8: How will the patient have arrested this once active lesion?

- Collagenous matrix has been denatured (irreparably damaged) by proteolytic enzymes intrinsic to the dentine itself (the zinc-dependent, acid-activated matrix metalloproteinases (MMPs)), as well as from bacteria, activated by those bacterial acids produced during the caries process.
- Bacterial load in this zone is very high.
- Dentine tubule structure is destroyed.

This zone of 'necrotic' dentine should be clinically removed when preparing a cavity, as it cannot be repaired histologically and it provides a poor-quality bonding substrate for adhesive materials to achieve an adequate seal, and inadequate physical restoration support.

- *Caries-affected dentine* (zones 2, 3, and 4 combined together; see Figure 1.14): the inner layer of carious dentine which can be repaired by the dentine–pulp complex, often distinguished as paler brown, harder, 'sticky and scratchy', or slightly leathery dentine (elicited by a sharp dental probe, which should not be used directly over the pulpal floor of a deep cavity).
  - Mineral dissolution still occurs, but to a lesser extent than in infected dentine, as the pH gradually rises towards the advancing front of the lesion and the ionic equilibrium begins to balance.
  - Collagen is still damaged by proteolysis, but to a lesser extent, so permitting dentine repair/mineral gain, as the proteinaceous scaffold for mineral crystal deposition now persists.
  - Bacterial load lessens, but there are still more anaerobic bacteria present.
  - Dentine tubule structure returns gradually within the depths of this zone.

The deepest layer of caries-affected dentine (zone 4 in Figure 1.14) can be described as hypermineralized translucent dentine (due to its glassy appearance in histological cross-section), one of several reparative reactions of the dentine–pulp complex to the caries process (see later). As can be seen in Figures 1.13 and 1.14, the lesion in dentine often has a dark brown discoloration within the caries-infected zone which then pales gradually through the depth of the lesion, towards the pulp. The aetiology of the colour changes is not clear, but a biochemical reaction between proteins and carbohydrates in a moist acidic biological environment, the *Maillard reaction*, may play a part. Not all lesions are uniformly dark brown; some rapidly advancing lesions may have a pale discoloration within the caries-infected zone, and there is no direct link between the colour of dentine and the bacteria present within these zones.

1.2.7 Carious pulp exposure

If the carious process cannot be modified by preventive or controlling measures and the lesion is not treated operatively in time with appropriate minimally invasive excavation techniques and a sealed adhesive restoration within the dentine, then the advancing front of the lesion will approach the dentine–pulp boundary. By this point, bacteria/toxins

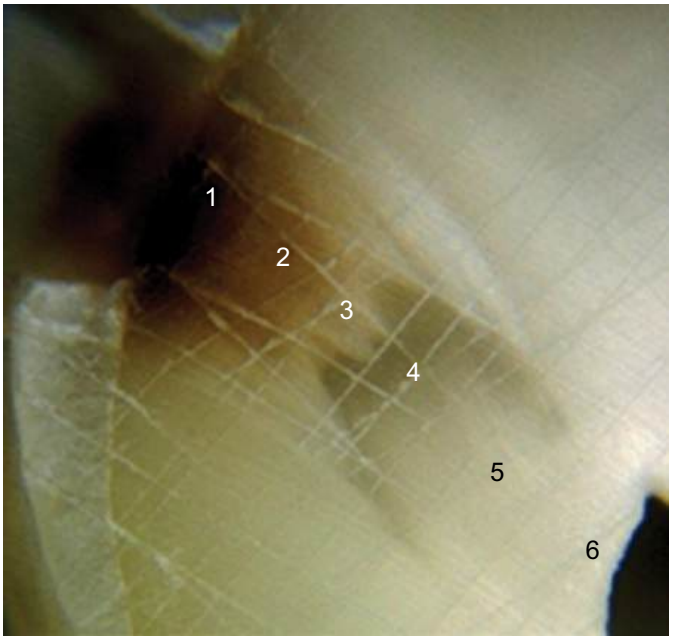


Figure 1.14 A mesio-distal section through a proximal lesion showing a cavity in the enamel and the histological colour changes through the dentine lesion (1, caries-infected dentine; 2, 3, and 4, caries-affected dentine; 4, translucent dentine; 5, 6, sound dentine). The surface criss-cross scratches were placed to act as reference markers during microscopy analysis of the sample.

Q1.9: What are the potential causes of the colour change in dentine caries?

will have penetrated into the pulp tissues causing an acute inflammatory response. Depending on the timescale over which this has happened, an initial acute pulpitic response (poorly localized short, sharp pain on exposure to hot, cold, or sweet stimuli) will evolve into a more chronic response, changing the symptoms experienced towards a dull, prolonged ache that may last for several minutes and is spontaneous and non-specific in origin (see Chapter 3). The cavity will probably have enlarged due to the undermined enamel having been broken away (including the marginal ridges of proximal lesions) during function, and will be noticeable to the patient as 'a hole in the tooth' or 'cavity.'

If the pulp chamber is breached physically by the lesion, a *carious exposure* may be created when excavating deep caries, and the exposed pulp tissue will bleed uncontrollably for several minutes before cotton wool pledgets can achieve haemostasis. In most cases of a carious pulp exposure, root canal treatment is the management option of choice (as long as the tooth is restorable with a worthwhile prognosis), but this will depend on the size of the exposure, the age of the patient (a young, well-vascularized pulp may have a better prognosis), and the severity of the carious process within the depth of the lesion (see Chapter 5). In rare cases, on late presentation, the pulp soft tissues undergo a hyperplastic cellular reaction and appear to herniate through the exposure, into the cavity.

If minimally invasive excavation principles and techniques are used to excavate carefully the deeper aspects of the carious lesion, the risk of a carious exposure is reduced (see Chapter 5). In cases where the pulp is deemed vital on initial clinical examination, it may be appropriate best practice to retain caries-affected dentine as an 'indirect pulp cap', sealing the cavity with a suitable long-term adhesive restoration and then reviewing the pulp status at subsequent planned recall appointments. Several clinical studies have shown this to be an effective minimally invasive method for maintaining the pulp vitality of teeth, so reducing the need for root canal treatment (see later).

1.2.8 Dentine–pulp complex reparative reactions

Dentine is a vital tissue containing the cytoplasmic extensions of odontoblasts, and must be considered together with the pulp, since the two tissues are so intimately connected. The dentine–pulp complex, like any other vital tissue in the body, is capable of defending itself. The state of the tissue at any time will depend on the balance between the attacking forces and the defence reactions. The defence reactions include deposition of *translucent dentine*, *tertiary dentine*, and *pulpal inflammation*.

Translucent dentine

Sometimes incorrectly referred to as 'sclerotic' dentine, this glassy zone of dentine (zone 4 in Figure 1.14) is caused by tubular infill with plate-like Whitlockite mineral crystals (β-octocalciumphosphate) at the advancing front of the lesion in an attempt to wall off the advancing lesion. Its appearance is due to the parity of refractive indices of intertubular and intratubular mineral, so allowing light to pass through the sectioned boundaries.

The Whitlockite deposits originate from a combination of a physico-chemical re-precipitation of calcium and phosphate ions diffusing towards the increasing pH environment of the lesion's deepest advancing front, and also possibly a vital process of new and rapid mineral deposition from the pulp via the odontoblasts. Even though hypermineralized, this zone of translucent dentine is softer than its deeper, sound counterpart due to the weaker crystalline orientation of Whitlockite than conventional hydroxyapatite crystals within the tubules (analogous to stacking dinner plates flat—too tall a stack and they topple over!) (see Figure 1.15).

Tertiary (reactionary/reparative/irritation/atubular) dentine

Secondary dentine is laid down throughout life as part of the normal pulp–dentine complex (see Figure 1.16). Tertiary dentine is laid down at the dentine–pulp border in response to a noxious stimulus (e.g. caries or those causing tooth wear), in an attempt to wall off and distance the pulp from the advancing noxious stimulus (see Figure 1.17). It may resemble secondary dentine histologically, but has an irregular tubular or atubular structure, depending on the speed of its creation. *Reactionary dentine* (see Figure 1.18) is deposited as a result of a mild irritant where original odontoblasts survive and are metabolically up-regulated. *Reparative dentine* (see Figure 1.19) is much more irregular and is deposited in response to a stronger irritant which compromises the vitality of the original odontoblasts. Progenitor cells from the sub-odontoblastic layer then differentiate and are up-regulated to produce an atubular defence reaction.

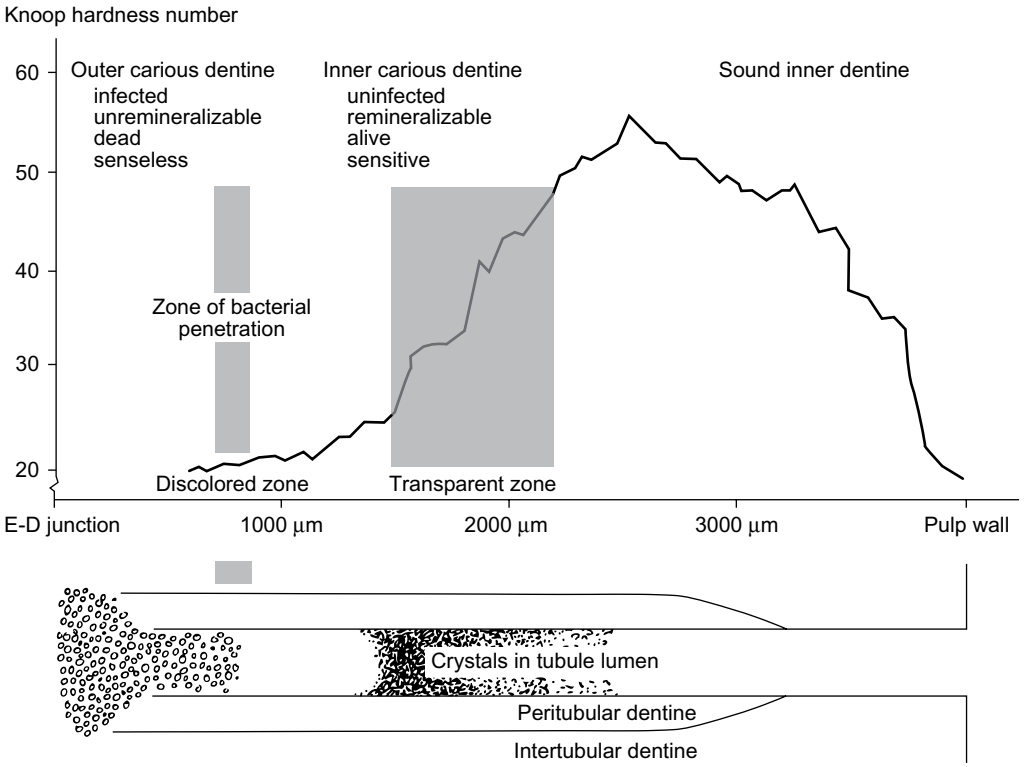


Figure 1.15 Diagram showing the changes in hardness of carious dentine (y-axis) from the enamel–dentine junction (EDJ) towards the pulp (x-axis), and relating this to the histological changes that occur through the dentine lesion. The transparent zone (synonymous with the translucent zone) is softer than the deeper, less mineralized, sound dentine. The diagram equates the bacterial content and mineral deposition within the tubule lumen through the progressive zones of carious dentine. Reproduced from O Fejerskov and E Kidd. *Dental Caries – the disease and its clinical management*. Copyright © 2008, John Wiley and Sons Ltd.)

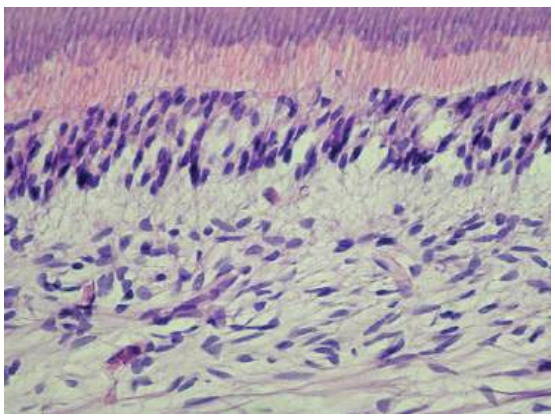


Figure 1.16 Haematoxylin and eosin (H&E) section of normal dentine–pulp complex with pale pink stained pre-dentine zone visible at the top of the image. This will undergo mineralization as part of the normal ageing process, as secondary dentine is laid down in the pulp chamber. (Courtesy of Carlos A. de Souza Costa.)

Q1.10: Can you identify two cellular zones in the pulp (the cell nuclei are stained purple)?



Figure 1.17 Right bitewing radiograph of a patient with high caries rate and multiple lesions. Note the dentine–pulp complex reparative response of LR6 to the distal dentine lesion—the distal pulp horn has been obliterated by deposits of tertiary dentine (arrow).

Q1.11: How many other carious lesions can you detect?

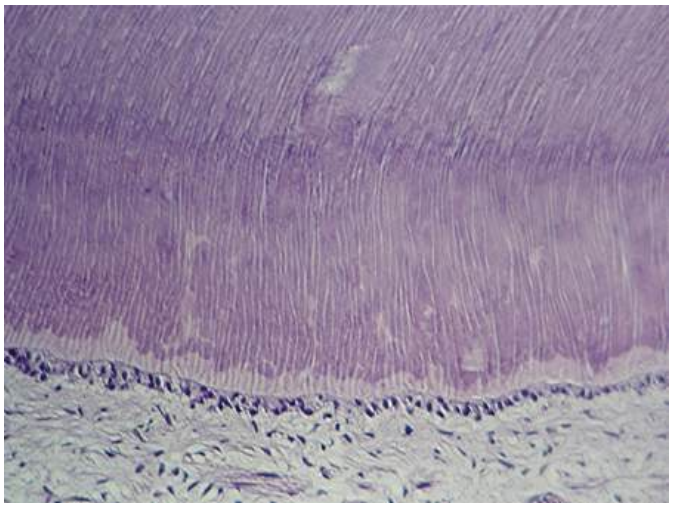


Figure 1.18 H&E section of pulp–dentine complex showing *reactionary dentine* in the middle of the field of view. (Courtesy of Carlos A. de Souza Costa.)

Q1.12: Can you identify five histological zones in the H&E section in Figure 1.18 from top to bottom?

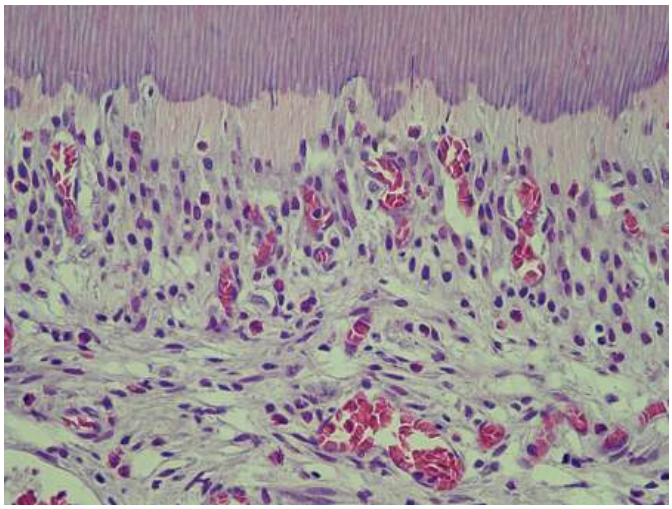


Figure 1.19 H&E section of pulp–dentine complex showing *reparative dentine*, along with the associated, disorganized, and newly differentiated odontoblasts. In the middle of the field of view can be seen bright red blood cells, associated with the pulp's inflammatory response. (Courtesy of Carlos A. de Souza Costa.)

Pulp inflammation

This is the fundamental response of all vascular connective tissues to injury. Inflammation of the pulp (pulpitis) may, as in any other tissue, be acute or chronic. In a slowly progressing carious lesion, toxins reaching the pulp may provoke chronic inflammation. However, once the organisms actually reach the pulp (a carious exposure), acute inflammation may supervene. Inflammatory reactions have vascular and cellular components. In chronic inflammation the cellular components predominate and there may be increased collagen production, leading to fibrosis, but without immediately endangering the vitality of the tooth. However, in acute inflammation the vascular changes predominate.

Infection is the most common cause of pulp inflammation, and caries is the most common microbial source. Dentine caries will result in pulp inflammation, and chronic inflammatory cells (macrophages, lymphocytes, and plasma cells) will infiltrate the pulp near the odontoblast layer. Indeed, this infiltration has been detected in response to initial enamel caries. This chronic inflammatory reaction is due mainly to the movement of bacterial toxins through the dentine tubules. Secretory immunoglobulins travel in the dentine fluid up the remaining patent tubules. With increasing carious involvement of enamel and dentine, the area of chronic inflammation increases in size, but it is believed to remain localized until pulp exposure. Bacteria may enter the pulp with polymorphonuclear leucocytes predominating, and acute inflammation can supervene, spread throughout the pulp, and result in necrosis.

1.3 Tooth wear ('tooth surface loss')

Tooth wear (TW) is 'the irreversible surface loss of dental hard tissues caused by factors other than caries or trauma.' It can be physiological, occurring slowly and naturally throughout life, or pathological, occurring at a much faster rate, usually caused by combinations of erosion, attrition, abrasion, and perhaps abfraction (see Table 1.2).

Erosion can be defined as the irreversible loss of dental hard tissues by a chemical process (acid attack) not involving bacteria. It is often the common denominator in the multifactorial aetiology of tooth wear. Sources of acid can be either intrinsic (stomach acid regurgitation) or extrinsic (dietary, environmental). Involuntary gastro-oesophageal reflux disease (GORD) is a common cause of intrinsic stomach acids entering the oral cavity on a regular basis, and occurs primarily due to transient relaxation or incompetence of the lower oesophageal sphincter. Certain factors in combination often predispose to GORD, and these factors may include:

- diet—fatty, spicy foods consumed in large quantities, especially late at night
- alcohol
- certain medications (e.g. diazepam)
- causes of increased gastric pressure, including obesity, pregnancy, posture (lying down increases pressure on the sphincter), and even excessive exercise
- gastro-oesophageal reflux predisposes to further GORD symptoms
- neuromuscular conditions.

Table 1.2 Tooth wear: common aetiology, features, and simple classification

Aetiological factors		Comments
Erosion	Intrinsic, regurgitation (GORD*, vomiting)	Common cause of erosion; affects palatal surfaces of maxillary anterior teeth, occlusal/buccal surfaces of lower molars. Stomach hydrochloric acid originates from: • involuntary GORD • involuntary/voluntary vomiting
	Extrinsic, dietary	Affects labial surfaces of maxillary anterior teeth. ↓ pH due to excess acidic food/drink intake: • citrus fruit and fruit juices • pickles, vinegar-containing foodstuffs • carbonated drinks (including diet/health drinks) • some mouthwashes have a low pH Drinking habits: through straw and frothing around mouth. Acids include citric, carbonic, acetic, hydrochloric, and phosphoric acids. Often associated with a healthy lifestyle—patient understanding is required to modify the erosive potential of the diet
	Extrinsic, environmental	Labial surfaces of maxillary anteriors/pitting. Rare nowadays due to stringent health and safety regulations in the workplace. Historically caused by industrial processes where acid was vaporized and inhaled (battery manufacturers, tanning factories)
Attrition	TW caused by occlusal tooth–tooth contact; occlusal facets match with opposing teeth; usually in combination with erosion. Often caused by grinding/parafunctional habits	
Abrasion	TW caused by tooth–non-tooth contact; hard toothbrushing with coarse toothpastes, dish-/V-shaped, smooth cervical lesions; incisal wear/grooves from long-term habitual behaviour (e.g. pipe smokers, milliners holding pins between their teeth, builders holding nails between their teeth, etc.)	
Abfraction	Cervical V-shaped enamel–dentine TW lesions with no history of abrasion. Aetiology not clear, but masticatory stresses may concentrate at cervical margins of teeth and perhaps open up pre-existing cracks/weaknesses in the tooth	

*GORD, gastro-oesophageal reflux disease.

Treatment for GORD will depend on the aetiology. As well as conservative management involving modifying diet, lifestyle, and the use of chewing gums, stomach acids can be neutralized using conventional antacids (e.g. Gaviscon), or their production limited with oral medications, including proton pump inhibitors (e.g. omeprazole (Losec)) or H_2antagonists (e.g. cimetidine). Surgical procedures can be performed to repair physical damage to the gastro-oesophageal system. Once this has been done, any dental damage can be repaired. The patient's dentist may often be the first to notice the problem through the dental manifestations, and appropriate referral to medical colleagues may be required.

Intrinsic acids can also enter the oral cavity through voluntary or involuntary vomiting, causes of which may be:

- psychosomatic:
  - eating disorders:
    (a) bulimia nervosa (affects 1–2% of the adolescent population; female:male ratio is 10:1)
    (b) anorexia nervosa (affects 0.1–1% of the teenage population; female:male ratio is 10:1)
  - rumination (voluntary regurgitation followed by re-digestion of stomach contents)
  - stress-induced psychogenic vomiting
- metabolic/endocrine:
  - pregnancy
- gastrointestinal disorders:
  - peptic ulcer/gastritis
  - hiatus hernia
  - achalasia—a condition associated with a narrowed lower oesophageal sphincter and reduced oesophageal motility, leading to stagnation and fermentation of ingested food within the oesophagus, and concomitant regurgitation
  - cerebral palsy
- drug induced:
  - primary—cytotoxic drugs
  - secondary—gastric irritation, alcohol, aspirin, and other non-steroidal anti-inflammatory medications.

Again, in all of the cases listed, the initial cause of the vomiting must be found out from the patient's history and examination, and the cause itself treated first, with close cooperation with the patient's medical practitioner, before restoring any damaged dentition (see Chapter 2).

The aetiology and features of abrasion and attrition have been outlined in Table 1.2. Patients suffering from dry mouth with reduced and/or chemically or physically altered saliva may also be more at risk from dental erosion. This is because they will lack the protective features of saliva, which include the ability to neutralize/buffer intra-oral acids and to supply mineral ions for potential mineral deposition, including statherins and proline-rich proteins. The mucin component of saliva may also help to protect the tooth surfaces from dietary acids, with contributions to the formation of the surface pellicle and biofilm. Erosion is often a contributory factor in the overall pattern of clinical tooth wear. Acid-softened tooth surfaces are more susceptible to long-term 'wear' forces from opposing teeth (attrition) or other external influences (abrasion). Clinical examples of these are shown in Chapter 2.

1.4 Dental trauma

Whereas caries and tooth wear are diseases of relatively slow onset, traumatic injuries are acquired suddenly, and when these involve the hard dental tissues and the pulp they usually require immediate operative management to stabilize the condition, provide pain relief, and restore function and appearance if possible. Trauma to the mouth can produce any combination of the following local injuries:

- lacerations to the lips, tongue, and buccal and gingival tissue
- alveolar fractures, so that a number of teeth become mobile within a block of bone
- complete or partial subluxation of a tooth
- root fracture
- damage to the apical blood vessels without fracture
- fracture of the crown of the tooth involving enamel alone, enamel and dentine, or exposure of the pulp (see Figure 2.25 in Chapter 2).

1.4.1 Aetiology

Trauma is commonly caused by the following:

- falls
- sports/athletics injuries
- blows from a heavy object
- fights
- car/bicycle accidents
- injuries sustained during convulsive seizures (e.g. epilepsy)
- battered child syndrome (the most difficult and yet the most important cause to diagnose).

Detection and management will be discussed in subsequent chapters. Examples of dental trauma can be seen in Figures 1.20 and 1.21.

In some cases, untreated traumatic injury can lead to the development of long-term pathology (see Figure 1.21).

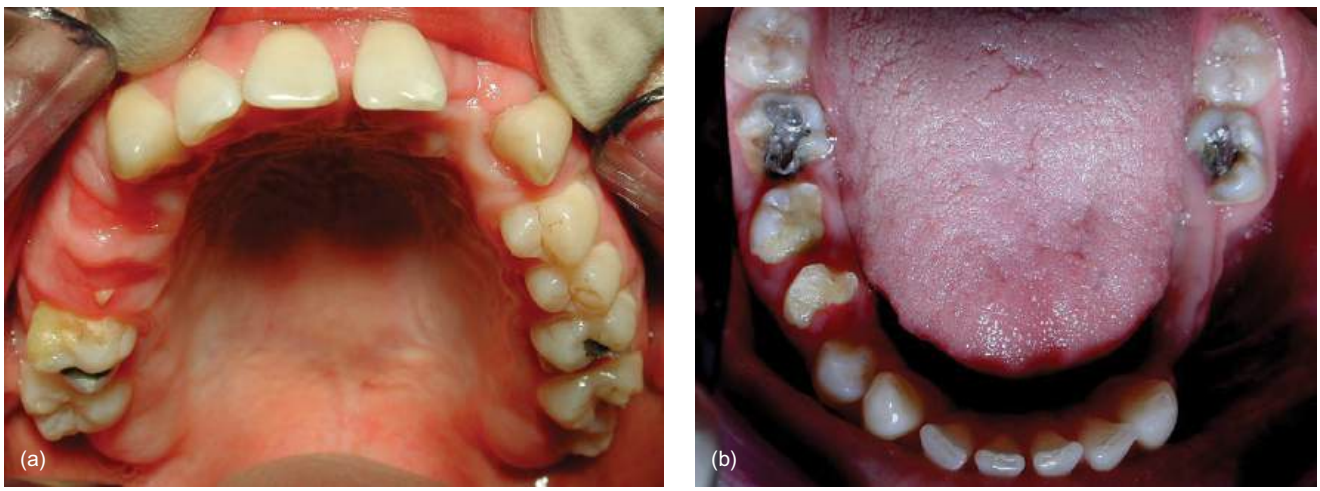


Figure 1.20 **a, b** Maxillary and mandibular occlusal views of a patient who has sustained multiple facial blows in a fight. Note the decoronated UR4,5 and large enamel and dentine fractures sustained on UR6, UL7, LR4,5,6, and LL6. UL2 and LL45 were avulsed in the incident.

Q1.13: What might have been the presenting dental complaints for the patient in Figure 1.20?



Figure 1.21 **(a)** A fit and well young adult has sustained an accidental blow to the teeth in the UL123 area several months earlier.

Q1.14: Can you detect any abnormal clinical findings from Figure 1.21 (clue—check the mucosae)?



Figure 1.21 **(b)** Periapical and **(c)** upper standard occlusal (USO) radiographs of the UL123 shown in **(a)**. Note the large well-demarcated pathological radiolucency originating from the root apex of the fractured and displaced root, post-core, and crown complex in the UL2. The lesion is tracking in the overlying mucosa to the swelling pointed out in **(a)**. **(d)** The extracted UL2 root with the post-core and crown removed, showing the crack sustained from the impact.

Q1.15: What other findings are evident from the USO radiograph?

1.5 Developmental defects

Teeth do not always develop normally, and there are a number of defects in tooth structure or shape which occur during development and become apparent on eruption. Such teeth are often unsightly or prone to excessive tooth wear or loss of clinical crowns, and thus they may require restoration to improve appearance or function or to protect the underlying tooth structure. These defects, their aetiology, and their clinical appearance are outlined in Table 2.7 in Chapter 2, with examples shown, and they include the acquired conditions of enamel hypoplasia, molar–incisor hypomineralization and intrinsic staining (fluorosis and tetracycline), as well as the hereditary conditions of hypodontia, amelogenesis imperfecta, and dentinogenesis imperfecta.

1.6 Suggested further reading and PubMed keywords

Fejerskov O and Kidd E A M (eds) (2008) *Dental Caries: the disease and its clinical management*. Oxford: Blackwell Munksgaard.
<http://eu.wiley.com/WileyCDA/WileyTitle/productCd1405138890,descCd-tableOfContents.html>

<www.ncbi.nlm.nih.gov/pubmed/advanced>

QR code image 1.1 Try searching the following keywords on PubMed for relevant further reading. You can access PubMed by scanning the QR code image or at the address

Keywords

Sjögren's syndrome; xerostomia; Maillard reaction and caries; dentine matrix metalloproteinases (MMPs); tooth wear; GORD and dental symptoms; hereditary dental conditions.

1.7 Answers to self-test questions

Q1.1: What differences in clinical appearance are there between the coronal and root surface caries lesion in Figure 1.3c, and how may they relate to lesion activity?

A: See Table 1.1.

Q1.2: What habit(s) may have contributed to this pattern of disease?

A: This child was allowed to suck a bottle of sweet drink frequently.

Q1.3: What aetiological factors may have contributed to this pattern of disease?

A: If you look closely at the lesion on LR3, you will notice an undermined periphery of enamel. Plaque stagnation has occurred beneath this rim, so activating the lesion metabolically.

Q1.4: Why might the lesion on the LR3 appear active whereas the adjacent lesion on the LR4 in the same patient appears arrested?

A: Improved oral hygiene would have arrested this active part of the lesion.

Q1.5: What ions have contributed to the creation of the intact surface zone and where have they come from?

A: Fluoride, calcium, and phosphate ions in particular help to form a more acid-resistant fluoride-substituted hydroxyapatite. These ions will have come from the saliva as well as from the biofilm, in higher concentrations, so creating a positive equilibrium with the tooth surface.

Q1.6: What features of the lesion in Figure 1.10 will help the dentist to conclude that it is active, how might these be detected, and how might the patient be managed?

A: The lesion has surface roughness (detectable as vibrations in the handle of the ball-ended explorer as it is gently run across the lesion surface) and was covered with plaque (detected visually with a disclosing agent). This patient was under preventive therapy, including modifications of their oral hygiene, possibly diet, and application of fluoride varnish to arrest the lesion.

Q1.7: How might this scenario be managed in a high caries risk patient?

A: Instruct the patient regarding their oral hygiene procedures, and if this does not improve then carry out debridement/air abrasion followed by application of a fissure sealant.

Q1.8: How will the patient have arrested this once active lesion?

A: They will have done so by judiciously removing the plaque biofilm regularly using proper oral hygiene methods, including floss. This combined with the use of a fluoride dentifrice, over time, will inactivate the incipient WSL into the brown spot lesion seen in the image.

Q1.9: What are the potential causes of the colour change in dentine caries?

A: Although not conclusive, this may be due to the Maillard reaction—a biochemical reaction between proteins and carbohydrates in a moist acidic biological environment.

Q1.10: Can you identify two cellular zones in the pulp (the cell nuclei are stained purple)?

A: Subjacent to the pre-dentine zone is the odontoblast layer with vertically oriented cells. Beneath this layer is the sub-odontoblastic cell-rich layer.

Q1.11: How many other carious lesions can you detect?

A: Distal/mesial (d/m) UR6, d/m UR5, m LR7, d/m LR6, d/m LR5, not including the grossly broken down UR4!

Q1.12: Can you identify five histological zones in the H&E section in Figure 1.18 from top to bottom?

A: Sound dentine, reactionary dentine, pre-dentine, odontoblast layer, and the relatively undifferentiated sub-odontoblastic cell-rich layer.

Q1.13: What might have been the presenting dental complaints for the patient in Figure 1.20?

A: Pain, difficulty chewing, sharp fractured teeth/fillings against the tongue or cheeks, not being able to bite properly, poor appearance, difficulty in brushing the teeth due to sensitivity, and tooth fractures.

Q1.14: Can you detect any abnormal clinical findings from Figure 1.21 (clue—check the mucosae)?

A: Difficult, but did you notice the mucosal swelling level with the muco-gingival junction adjacent to the UL3?

Q1.15: What other findings are evident from the USO radiograph?

A: The near complete obliteration of the pulp chamber and root canal spaces in the UL1.